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CLINICAL PRACTICE UPDATES

AGA Clinical Practice Update on Sclerosing Mesenteritis: Commentary

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DESCRIPTION: **The purpose of this American Gastroenterological Association Institute Clinical Practice Update is to review the available evidence for diagnosing and treating, as well as examine opportunities for future research in, sclerosing mesenteritis.**

METHODS: **This Clinical Practice Update expert commentary was commissioned and approved by the AGA Institute Clinical Practice Updates Committee and the AGA Governing Board to provide timely guidance on a topic of high clinical importance to the AGA membership, and underwent internal peer review by the Clinical Practice Updates Committee and external peer review through standard procedures of *Clinical Gastroenterology and Hepatology*. This expert commentary incorporates important and recently published studies in this field, and it reflects the experiences of the authors who are gastroenterologists with expertise in this topic.**

Keywords: Sclerosing Mesenteritis; Mesenteric Panniculitis; Mesentery; Small Bowel; Computed Tomography; Abdominal Pain.

Sclerosing mesenteritis (SM) is an uncommon disorder of the bowel mesentery characterized by inflammation, fat necrosis, and fibrosis that was first described 100 years ago.[1] However, the etiopathophysiology of this disorder is still poorly understood. Because of the widespread use of computed tomography (CT) imaging in contemporary medicine to investigate abdominal pain, the diagnosis of SM (or findings suspicious for SM) is increasingly common. Therefore, gastroenterologists need to be cognizant of this diagnosis and its management.

Nomenclature

Various terms have been used for SM over the years including mesenteric panniculitis, retractile mesenteritis, “misty mesentery,” mesenteric lipodystrophy, liposclerotic mesenteritis, mesenteric Weber-Christian disease, xanthogranulomatous mesenteritis, idiopathic mesenteric fibrosis, inflammatory pseudotumor, and systemic nodular panniculitis. These terms likely represent a continuum of the same disease with differing ratios of fibrosis, fat necrosis, and inflammation, and thus the unifying term of “sclerosing mesenteritis” has been proposed.[2]

Epidemiology

Retrospective CT studies report a frequency of SM findings ranging from 0.6% to 1.1%.[3] Demographic data are limited, but a large case series reported that patients with SM had a mean age of 55 years, more likely to be male and White race.[4]

Clinical Features

Signs and Symptoms

Up to 60% of patients with SM findings on CT are asymptomatic,[5] but this is likely an underestimate, given that asymptomatic persons do not all undergo imaging. Among those with symptoms, abdominal pain is most common (up to 75%), with bloating (10%–25%), diarrhea or constipation (10%–25%), nausea and vomiting (5%–15%), and weight loss (20%) also frequent.[6] A palpable abdominal mass is often not found.

Patients with SM do not have a higher prevalence of autoimmunity, but may have increased rates of metabolic syndrome, obesity, coronary artery disease, and urolithiasis.[7]

Abbreviations used in this paper: CT, computed tomography; PET, positron-emission tomography; SM, sclerosing mesenteritis.

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Imaging

CT diagnostic criteria for SM were established by Coulier.[8] These are: (1) a hazy mesenteric fat mass displacing intestine; (2) higher attenuation in the mesenteric fat mass than adjacent mesentery, but no enhancement with intravenous contrast; (3) the presence of lymph nodes <1 cm;[4] (4) a hypoattenuated halo sign around central vessels; and (5) a thin surrounding fibrotic pseudocapsule. CT images with these diagnostic features are shown in Figure 1.

The presence of 3 or more of these features is suggestive of SM. A hazy mesentery without nodes, halo sign, or pseudocapsule is therefore not diagnostic for SM and, without other symptoms, typically does not require follow-up. Although definitive diagnosis requires biopsy, in patients with characteristic findings on imaging and no alarm features (fever, night sweats, unintentional weight loss), the risk of malignancy is very low and a presumptive diagnosis can be made without a biopsy in most cases.[7] Calcifications and retraction of the surrounding bowel, characteristic of more severe disease and possible malignancy (Figure 1*B* and *C*), typically warrants biopsy, unless an association with IgG4 disease can be established (discussed later).

Laboratory Findings

Erythrocyte sedimentation rate and C-reactive protein may be elevated, but not in all cases. There is scant literature on these biomarkers and their role in management or follow-up is uncertain at this time.

IgG4

Rarely, patients with SM have IgG4-related disease, which often has an atypical presentation.[5,9] On CT, the inflammation can appear unlike SM, with malignant features (indistinct margins, apparent invasion of adjacent organs, multiple enlarged lymph nodes). Only half have elevated serum IgG4 levels to suggest the diagnosis.[10] A surgical biopsy with IgG4 staining may be required if the IgG4 level is normal and the imaging is not diagnostic for SM.

Severe Disease

SM features suggesting a more severe clinical course include: (1) omental, retroperitoneal, pelvic fat, or colonic mesentery involvement;[2] (2) associated bowel obstruction/ischemia, venous thrombosis/thromboembolism, sepsis, protein losing enteropathy, or renal failure;[6,11] (3) marked C-reactive protein or erythrocyte sedimentation rate elevations;[12] (4) coexisting autoimmune disease;[12] (5) arterial narrowing; and/or (6) ascites.

Differential Diagnosis

The differential diagnosis for SM includes other mesenteric conditions, usually distinctly different on imaging.

Non-Hodgkin lymphoma can present with mesenteric infiltration or a mass lesion that is difficult to distinguish from SM on imaging alone. It also can cause enlargement of lymph nodes in the mesentery, with or without enlarged nonmesenteric lymph nodes or splenomegaly.[13] Characteristic imaging features of SM-associated lymphoma are noted next.

Peritoneal carcinomatosis is usually associated with ascites and multiple enhancing mesenteric, peritoneal, and/or omental nodules, although occasionally a poorly defined infiltrating mass is mistaken for SM. A desmoplastic reaction may occur with some mesenteric malignancies, with fibrotic retractions and deformity.

Carcinoid tumor is often associated with focal bowel and liver lesions,[14] but atypical cases with only mesenteric lesions have been seen.

Mesenteric fibromatosis (desmoid tumors) are tumor-like fibromatoses, commonly seen in familial adenomatous polyposis.[15] Desmoid tumors tend to be multifocal, have a noninflamed, mass-like and/or infiltrative appearance in the mesentery, and can encase vessels and intestine. Magnetic resonance imaging is the preferred diagnostic test.[16]

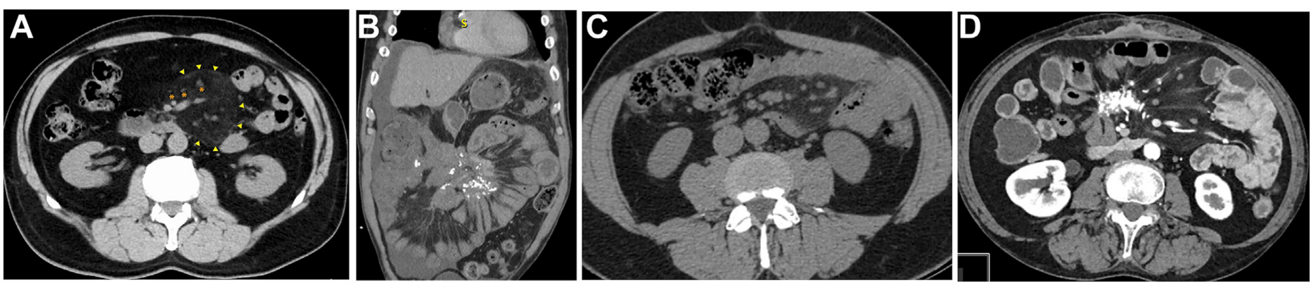

Figure 1. CT images of patients with sclerosing mesenteritis. (*A*) CT with classic diagnostic findings as described in the text, including central vessels with halos (*asterisks*) and surrounding pseudocapsule (*arrows*). (*B*) Calcified mesenteric mass with bowel retraction. (*C*) “Misty mesentery” in a patient with SM. (*D*) Calcified mesenteric mass with retraction and bowel thickening.

Encapsulating peritoneal sclerosis most commonly occurs in peritoneal dialysis patients, although it can be idiopathic, and is characterized by small bowel loops that are encased in a thickened peritoneum, often with dilated proximal bowel. A cauliflower appearance of compressed bowel loops is a classic finding of encapsulating peritoneal sclerosis, with the encasing and contrast-enhancing peritoneum creating a "cocoon" appearance. Pockets of ascites occur in 90%, and peritoneal calcifications in 70%. Intraperitoneal strands may cause the thickened bowel wall to appear trilaminar.[17,18]

Retroperitoneal fibrosis is a soft tissue fibrotic mass from the distal aorta between L4 and L5 to the iliac arteries, encasing the ureters. Hydronephrosis and medial deviation of the proximal two-thirds of the ureters are common.[19] Symptoms usually include lower back or flank pain, distinct from the focal abdominal pain seen in SM. Retroperitoneal fibrosis can look like SM histologically, and patients can have both processes.

Pseudomyxoma peritonei is a progressive intraperitoneal accumulation of mucinous ascites caused by a mucin-producing neoplasm (eg, appendiceal or ovarian tumors), with fluid loculated along peritoneal surfaces. This classically results in a scalloped appearance of coated organs.[20]

Other mesenteric processes that usually do not have the typical appearance of SM include gastrointestinal stromal tumors, sarcomas, mesenteric edema (hypoalbuminemia, cirrhosis, congestive heart failure, trauma) and certain infections (mycobacteria, cryptococcus, schistosomiasis, HIV, and cholera).[6,21]

Relationship of Sclerosing Mesenteritis to Cancer

A frequent concern of patients and providers regarding SM is whether the imaging findings portend an increased risk of intra-abdominal cancer. Although some have suggested that SM may be a paraneoplastic syndrome,[5,9] current evidence does not support this.[6,22–24] However, there does seem to be an overlap with SM findings and certain types of cancer, including lymphoma,[13,25] gynecologic malignancies, and carcinoid tumors.

The presence of a mesenteric mass or intra-abdominal lymph node >10 mm in size should raise suspicion for an underlying malignancy.[13,e1] In 1 prospective study of more than 400 patients with SM, the rate of subsequent malignancy was 1%; all were B-cell lymphomas, and had either a soft tissue nodule >10 mm or abdominopelvic lymphadenopathy.[13] Use of positron-emission tomography (PET) has been investigated in these cases, with favorable diagnostic performance.[25,e2]

Of note, SM (often asymptomatic) has also been reported following cancer therapy with immune checkpoint inhibitors, and is generally responsive to steroids.[e3]

Management

Management is based on presentation, with treatment generally reserved for symptomatic patients (Figure 2). Fever should prompt evaluation for infection before attribution to SM.

Biopsy

If the imaging is classic for SM (Figure 1) and there are no concerns for cancer, then biopsy is unnecessary before initiating treatment.

Biopsy should be considered if there are SM features concerning for malignancy, including: (1) associated lymph nodes >1 cm;[13,e1,e2] (2) lymph nodes with SUV_{max} $\geq$3.0 on ^{18}FDG PET/CT;[e2] (3) lymphadenopathy in another intra-abdominal location;[13,e1] and/or (4) spiculated, necrotic, calcified, or invasive appearance, including absence of central halo sign.[13]

In patients with features concerning for malignancy, ^{18}FDG-PET/CT is helpful for characterization and to identify all potentially accessible nodes for biopsy.

IgG4-related SM disease can appear invasive and have accompanying lymphadenopathy. Biopsy is generally required to make this diagnosis (and to exclude malignancy).

Follow-Up Imaging

Published recommendations for follow-up imaging because of concerns about associated malignancy vary widely, from no repeat imaging,[13,e1] to annual repeating examinations up to 5 years.[9,e4] In a meta-analysis of 4 case-control studies, an increased risk of cancer was not detectable,[24] but another study reported a small long-term increased risk of lymphoma in those without initial features of malignancy, with 80% detectable in the first year.[13] The second reason for repeat imaging is to assess progression of inflammation. Although the data are limited, except in the most severe cases, most patients—requiring treatment or not—achieve stable disease without radiologic progression after 2 years.[e5] During follow-up imaging, if any SM features concerning for malignancy occur, including significant growth in the mesenteric mass or new lymphadenopathy, biopsy should be performed.[25]

For asymptomatic cases without initial features concerning for associated malignancy, we suggest that patients be followed with annual CT scans for 2 years to ensure stability, and thereafter performed for concerning symptoms.

Medical Therapy

Therapy of SM is generally aimed toward symptom relief, because there is limited evidence that treatment can improve imaging or prevent complications.

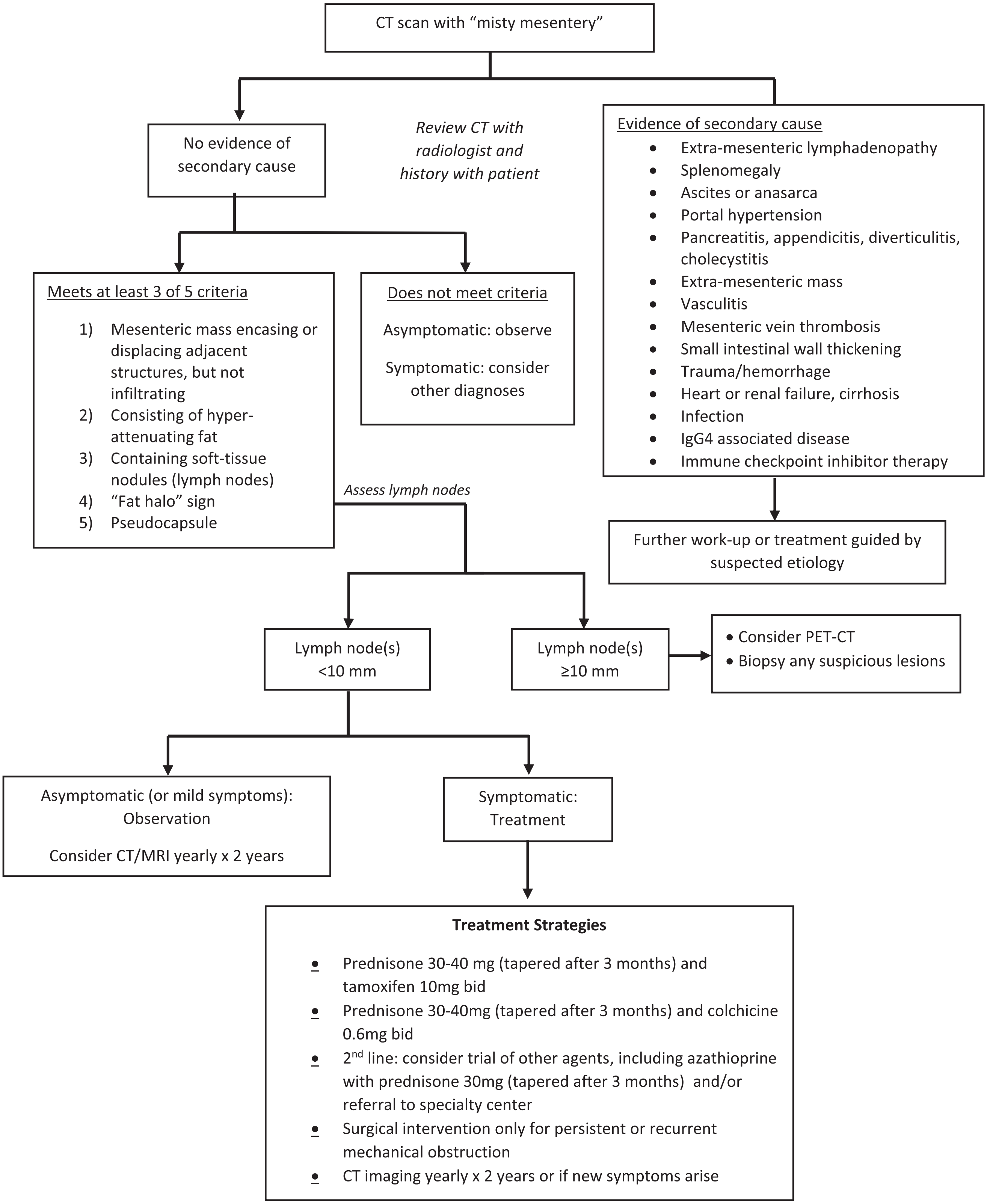

Figure 2. Algorithm for the evaluation and treatment of patients with sclerosing mesenteritis. MRI, magnetic resonance imaging.

Accordingly, patients without symptoms are usually not treated. In several studies, only 1%–6% of patients who presented with imaging findings of SM required treatment.[4,21,e6] However, this figure may be higher in referral centers; in 2 large observational studies, approximately 50% received treatment initially, with more requiring treatment over time.[e5,e7] In our experience, most patients with SM are referred to gastroenterologists, and checkpoint inhibitor–associated SM is divided between gastroenterologists and oncologists.

Because the immune response is unknown and controlled trials are unavailable, drug selection in SM is based on open-label studies or case reports. Our suggested approach is shown in Figure 2. General measures include treating concomitant gastrointestinal disorders if present, such as constipation, gastroesophageal reflux disease, and/or irritable bowel syndrome. One of our coauthors (DSP) routinely uses tricyclic antidepressants for the mild pain associated with SM.[e8]

Corticosteroids

Prednisone monotherapy (30–40 mg a day) may improve short-term symptoms,[e7] but long-term treatment is not advised because of side effects. In a large retrospective series of prednisone-treated patients, treatment was typically started at 40 mg a day, continued for 3–4 months, then tapered 5 mg/week.[e5] Any benefit is typically lost after drug discontinuation.

Tamoxifen

The most commonly used therapy is tamoxifen, 10 mg twice daily, alone or together with prednisone 30–40 mg daily.[e7] Initially chosen for its antifibrotic activity against transforming growth factor-β because mesenteric fibrosis is the hallmark of more advanced disease,[e7] the actual tamoxifen mechanism is unknown. With combination therapy, prednisone is usually tapered after 3 months, and tamoxifen continued for maintenance, although how long to continue has not been determined.

Maximal response to prednisone and tamoxifen occurs around 4–6 months, with 50%–60% of patients responding.[e5,e7] Tamoxifen side effects include antiestrogenic symptoms (hot flashes, sweating, vaginal bleeding), alopecia, fatigue, thrombosis, and laboratory abnormalities (thrombocytopenia, elevated liver enzymes). In responders with mild tamoxifen side effects, 10 mg daily may maintain response and improve tolerance. Contraindications to tamoxifen include a history of venous thrombosis, pulmonary embolism, or significant cerebrovascular, cardiovascular, or peripheral vascular disease. Active tobacco use is considered a relative contraindication.

Colchicine

Colchicine with prednisone has been used successfully, with a response rate of around 50% in a small series.[e5,e7,e9] Colchicine is prescribed at 0.6 mg twice a day with prednisone 40 mg daily, tapering prednisone after 3 months.[e5] Gastrointestinal side effects may limit colchicine use, and is contraindicated in renal insufficiency.

Azathioprine

Azathioprine is typically dosed similar to inflammatory bowel disease (2–2.5 mg/kg/d) in patients who are normal thiopurine methyltransferase metabolizers.[12,e10] A major limitation of azathioprine is a slow onset of action, so coadministration with prednisone is used early in therapy.

Other Immunosuppressants

Drugs with limited data include rituximab,[e5,e11] thalidomide,[e12] cyclophosphamide,[e10,e13] methotrexate,[e7,e14] infliximab,[e15] and ustekinumab.[e16] All of these agents have the potential for significant side effects, and thus are typically reserved as second-line therapies for severe, refractory disease.

Management of Severe Disease Complications

Bowel obstruction is managed with nonoperative management when feasible, but if unsuccessful, surgical bypass may be required.[e17–e21] In most, complete resection is impossible because of vascular compromise and disease extent.

Patients with chronic ischemia caused by mesenteric vascular obstruction are considered for revascularization. Few are candidates for endovascular revascularization because of the long length or distal location of the vessel. Surgical mesenteric revascularization is rarely feasible because of the extensive inflammatory disease. Radiation therapy has had equivocal success.[e22,e23]

Surgery does not cure SM. In 1 study, only 10% of patients who underwent surgery required no additional treatment.[e7] Recurrent disease can be detected by radiologic findings or new clinical symptoms. In patients with persistent or recurrent symptoms following surgery, we suggest medical management similar to patients with nonobstructive symptoms.

Conclusions and Future Directions

In summary, SM is an idiopathic autoimmune disease of the mesenteric fat, increasingly found because of widespread use of CT imaging, and forms a continuum that includes the terms “misty mesentery” and mesenteric panniculitis. Most cases are asymptomatic, do not require treatment, and can be followed. When symptomatic, the abdominal pain on examination should correspond to the SM lesion on imaging and is treated with anti-inflammatory medications tailored to disease severity and clinical response. Biopsy is not required if the SM meets 3 of the 5 CT criteria reported by Coulier[8] (discussed previously) and lacks features of more aggressive disease or malignancy. There is no surgical cure. Checkpoint inhibitor–associated SM is a unique case that responds readily to steroids and confirms the autoimmune nature of the condition. The most severe SM cases should be considered for referral to tertiary centers with established expertise.

Current treatments are largely corticosteroid-based, often requiring months to achieve a clinical response. Because the etiopathogenesis is unknown and treatment protocols have been empirically derived, future investigation for symptomatic SM should focus on the nature of the inflammatory response, including causative cytokines and other proinflammatory mediators, the goal being targeted therapy with fewer side effects and a more rapid clinical response.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Clinical Gastroenterology and Hepatology* at www.cghjournal.org, and at https://doi.org/10.1016/j.cgh.2024.12.025.

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Conflicts of interest
The authors disclose the following: Mark Worthington received renumeration from TriCity Surgery Center, Prescott, AZ. Jacqueline Wolf received renumeration from AbbVie Inc, Align Technology, Alnylam Pharmaceuticals Inc, CVS Health ORP, Dexcom Inc, Exact Sciences Corp, HCA Healthcare Inc, Johnson & Johnson, Eli Lilly & Co, McKesson Corp, Moderna Inc, Regeneron Pharmaceuticals Inc, Sarepta Therapeutics Inc, Seagen Inc, Stryker Corp, and Thermo Fisher Scientific. Seth Crockett served as a consultant for IngenioRx. Darrel Pardi served as a consultant for Boehringer Ingelheim; and received research support from Atlantic, ExeGI Pharma, Rise Therapeutics, Janssen, Pfizer, Seres, Applied Molecular Transport, Takeda Pharmaceuticals, and Vedanta Biosciences.

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